



Differentiable Dynamics for Autonomous Micro-Mobility Navigation

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Motivation

- Autonomous micro-mobility vehicles (MMVs) — wheelchairs, scooters, bicycles — must move predictably in pedestrian-shared spaces. Poorly modeled dynamics cause over-/under-steering and abrupt stops.
- Differentiable AV stacks such as DiffStack rely on unicycle or kinematic bicycle models: **no tire slip, friction, load transfer, or lean**.
- GM3** [Cai et al., ICRA '26] models tire-level MMV dynamics for arbitrary wheel layouts, but is **non-differentiable** — it cannot be trained inside a gradient-based stack.

Contributions

- DiffKBM** — the classical kinematic bicycle model, fully differentiable, with learnable wheelbase L .
- DiffGM3** — a differentiable reformulation of GM3: smoothed brush-tire forces, singularity-free steering, lean as a second-order DOF.
- Learns physical parameters that are hard to measure: friction, contact length, stiffness, inertias, damping.
- Drop-in dynamics layer for open-loop imitation (DiffStack) and closed-loop planning (DiffMPC).

Making GM3 Differentiable

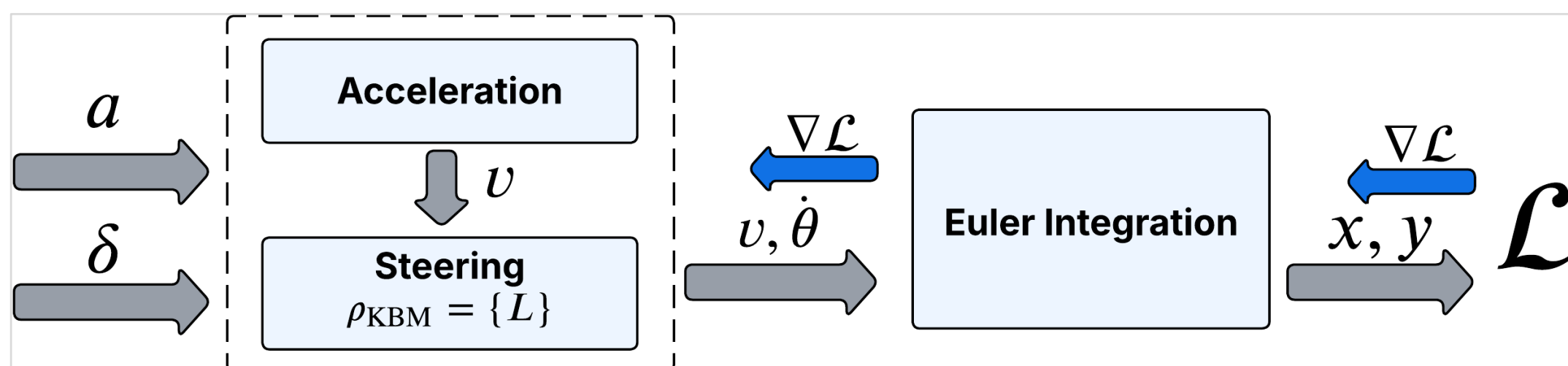
GM3's brush-model forces switch hard at the adhesion–sliding boundary. Every discontinuity is replaced by a smooth surrogate so gradients flow.

Discontinuity	Smooth surrogate
Adhesion / sliding switch at $\sigma = 1/\psi$	sigmoid gate $G = 1 / (1 + e^{-k(\sigma - 1/\psi)})$
$\text{sgn}(\alpha)$ in F_y and ψ^*	$\tanh(k\alpha)$
$ \sigma $ at zero slip	$\sqrt{(\sigma x^2 + \sigma y^2 + \epsilon)}$
Ackermann singularity at $\delta = 0$	$\delta_i = \arctan((L \tan \delta) / (L - y_i \tan \delta))$
Hard parameter / control limits	$\tanh$ soft clamp (implicit prior)

Worst-case force error decays as $(1/k)^3$ — and receding-horizon MPC re-plans from the measured state every step, so the smoothing mismatch cannot accumulate.

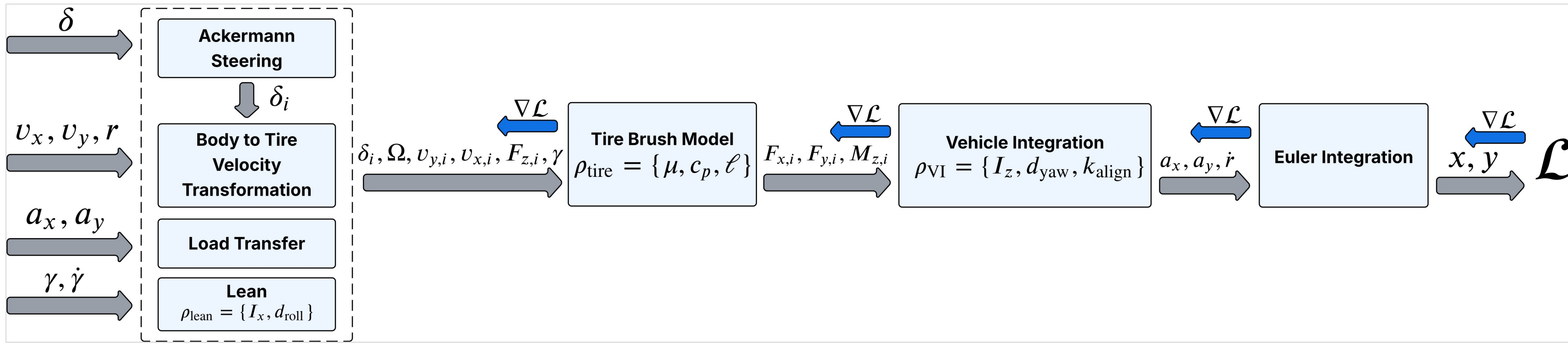
DiffKBM Baseline

Euler-discretized bicycle kinematics, differentiable almost everywhere, so wheelbase L is learnable. **No slip, lean, or load transfer** → no way to damp accumulated heading error.



DiffKBM: the wheelbase L enters the steering term; θ and v are Euler-integrated to a pose, and gradients reach L .

DiffGM3: Architecture



Fully differentiable pipeline: the steering input δ is mapped to per-tire angles; tire velocities, load transfer, and lean feed the smoothed brush model; per-tire forces integrate to vehicle acceleration and an Euler-integrated pose. Gradients reach every module, so ρ_{lean} , ρ_{tire} , and ρ_{VI} train directly on MMV trajectory data.

Steering, Yaw & Lean

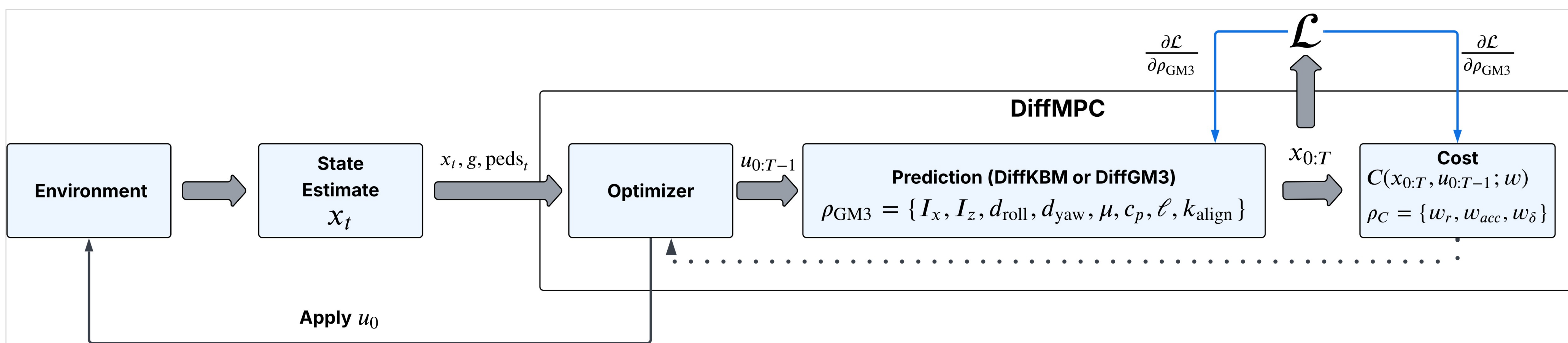
- Smooth Ackermann** removes the singularity at $\delta = 0$; steering is soft-clamped to $|\delta| \leq \delta_{\text{max}}$.
- Yaw damping** $\dot{r} = Mz / Iz - d_{\text{yaw}} \cdot r$ corrects the oversteer KBM cannot.
- Lean as a real DOF**, second order: $I_x \ddot{\gamma} + d_{\text{roll}} \dot{\gamma} + m g h_{\text{cg}} \sin \gamma = m a_y h_{\text{cg}} \cos \gamma$.
- $\partial y_{\text{out}} / \partial y_{\text{in}}$ never vanishes, so the optimizer recovers even if the vehicle “falls over” during an epoch.

Learnable Parameters

Component	Optimized parameters
Vehicle body	$k_{\text{align}}, d_{\text{yaw}}, d_{\text{roll}}, I_z, I_x$
Tires (each wheel)	μ, c_p, ℓ
Kinematics (DiffKBM)	L

Calibrated on 30 IMPTC trajectories with DiffMPC-inferred controls, since no MMV dataset records steering angle and wheel speed. Soft clamping keeps the weakly excited friction parameters physically feasible.

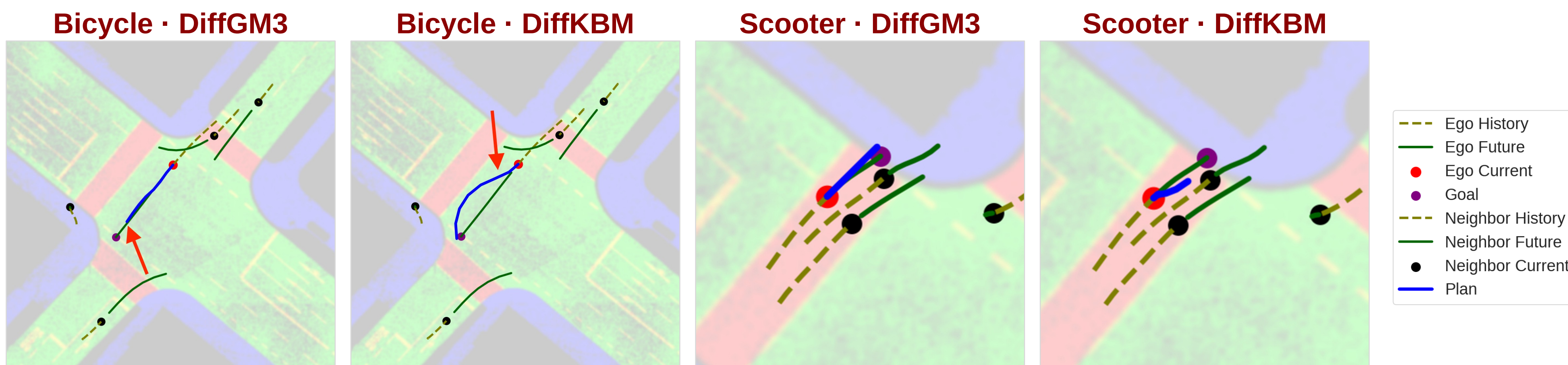
Closed-loop: DiffMPC + DiffGM3



At each timestep: the optimizer rolls candidate controls through DiffKBM or DiffGM3, scores them with a differentiable cost (goal tracking, turning-circle CBF, control regularization), and applies u_0 in a receding-horizon loop. Gradients flow through both the dynamics and the cost.

Open-loop Trajectory Matching (IMPTC)

DiffKBM and DiffGM3 replace the dynamically-extended unicycle model inside DiffStack and are trained to reproduce real intersection trajectories — 617 bicycle, 233 motorcycle, and 85 scooter tracks.



DiffGM3 stays on the ground-truth path because damping and align gains prevent overturning; DiffKBM reaches the goal but veers off-path (bicycle) or turns into another agent and stalls (scooter).

Open-loop Results

Metric	Mode	DiffGM3	DiffKBM	Δ
ADE (m) ↓	Bicycle	2.06	2.08	−1.2%
	Scooter	1.45	1.61	−9.8%
	Motorcycle	1.74	1.78	−2.4%
NLL ↓	Bicycle	1.07	1.10	−3.0%
	Scooter	1.23	1.28	−2.1%
Plan loss ↓	Motorcycle	3.02	3.17	−4.8%
	Bicycle	149.1	155.6	−4.2%
	Scooter	142.8	119.6	+19.4%
	Motorcycle	212.2	298.5	−28.9%

DiffGM3 wins on ADE and NLL for every mode. The scooter planning-loss regression comes from DiffGM3 braking before the goal — it holds the path while DiffKBM swerves off and turns back.

Closed-loop Results

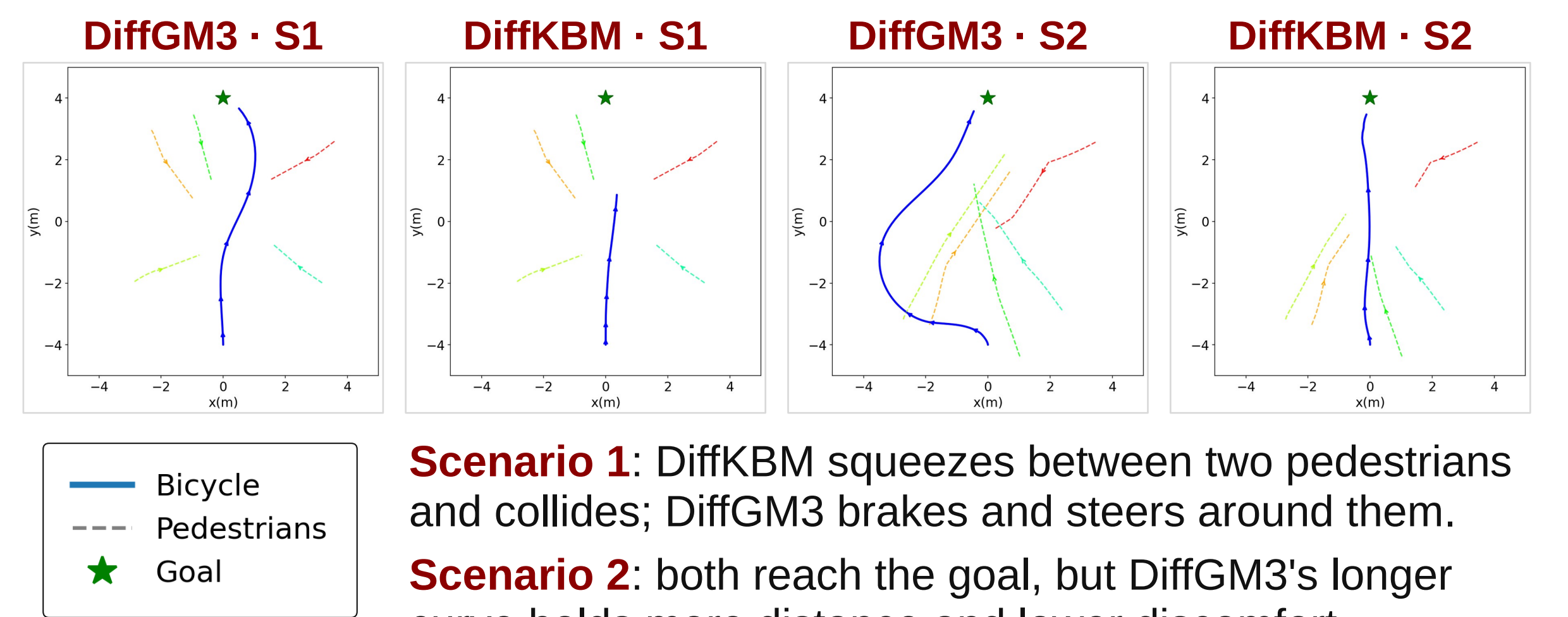
100 CrowdNav pedestrian scenarios; only the dynamics model differs.

Model	Succ ↑	Coll ↓	Inc. ↓	Nav (s) ↓	Disc ↓	Dist ↑
DiffKBM Bicycle	0.62	0.38	0.00	4.02	0.08	0.13
DiffGM3 Bicycle	0.64	0.17	0.19	5.13	0.02	0.12
DiffGM3 Scooter	0.63	0.13	0.23	6.14	0.03	0.13
DiffGM3 Wheelchair	0.60	0.23	0.17	6.30	0.03	0.12
DiffGM3 Tricycle	0.67	0.23	0.17	4.82	0.04	0.11

vs. DiffKBM bicycle: 55% fewer collisions, 75% lower discomfort frequency, +3% success rate.

DiffGM3's longer navigation times reflect veering around pedestrians; DiffKBM's faster times are partly collisions ending runs early.

Closed-loop Scenarios



Conclusion

- A unified tire-level MMV model can be made fully differentiable with bounded approximation error.
- DiffGM3 reproduces real trajectories more accurately than DiffKBM and navigates crowds far more safely — from one formulation shared by bicycles, scooters, tricycles, and wheelchairs.
- Next: high-curvature and long-horizon data, more platforms, differentiable simulation, and hardware deployment.

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